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Paleomagnetism of the Miocene intrusive suite of Kidd Creek: Timing of deformation in the Cascade arc, southern Washington

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Abstract. Paleomagnetic study of the intrusive suite of Kidd Creek in the southern Washington Cascades (23 sites in dikes and sills) was undertaken to help determine if these rocks are comagmatic and whether they postdate regional folding of the volcanic arc. Fission track and ^{40}Ar-^{39}Ar age determinations indicate an age of ~12.7 Ma (middle Miocene) for these rocks. The similarity of normal-polarity characteristic directions for most samples corroborate the available geochemical data indicating that these rocks are most likely comagmatic. Reversed-polarity directions for samples from four sites, however, show that emplacement of Kidd Creek intrusions spanned at least one reversal of the geomagnetic field. The paleomagnetic directions for the dikes and sills fail a fold test at the 99% confidence level indicating that the Kidd Creek rocks postdate regional folding. The mean in situ direction also indicates that the Kidd Creek and older rocks have been rotated 22° ± 6° clockwise about a vertical or near-vertical axis from the expected Miocene direction. Compression and regional folding of the Cascade arc in southern Washington therefore had ended by ~12 Ma prior to the onset of deformation resulting in rotation of these rocks.

1. Introduction

The middle Miocene intrusive suite of Kidd Creek represents the youngest Tertiary magmatic event known in an area on the western flank of the southern Washington Cascades, located between the Cowlitz and Lewis Rivers (Figure 1), recently mapped by *Swanson* [1991, 1992, 1993]. The Kidd Creek suite consists of sills and dikes of hornblende diorite, microdiorite, and their finer-grained equivalents. Major and trace element geochemical data indicate that these rocks are likely comagmatic [*Marso and Swanson*, 1992]. In general, dikes trend radially to the subvolcanic, mineralized, hornblende diorite intrusion of McCoy Creek (Figure 2) and have subvertical attitudes in Oligocene to lower Miocene volcaniclastic and volcanic host rocks. Regional dip angles for the host rocks are as high as 40°. The consistent map pattern and uniform attitudes of the dikes suggest that the intrusive suite of Kidd Creek is postfolding [*Swanson*, 1992]. In some field localities, near-vertical dikes can be interpreted to merge with dipping sills or to twist into bedding planes and become sills. If this interpretation is correct, the sills are also postfolding.

Broad open folds with north to northwest trending axes are the dominant structure of the southern Washington Cascades [*Hammond*, 1980; *Walsh et al.*, 1987], but the age of this folding is poorly known. Most of it likely occurred in middle to late Miocene time, because mid-Tertiary rocks are affected, whereas those of Pliocene age apparently are not [*Swanson et al.*, 1989]. Paleomagnetic investigation of the intrusive suite of Kidd Creek was undertaken to test the following two hypotheses: (1) that the dikes and sills are comagmatic and (2) that the Kidd Creek suite postdates regional folding of the Cascade arc in southern Washington. Fission track and ^{40}Ar-^{39}Ar analyses were also made to determine an accurate age for the intrusive suite of Kidd Creek.

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2. Geologic Setting

The Cascade magmatic arc in Washington has been active since the earliest Oligocene (~36 Ma) and consists of a north-south trending belt of Oligocene and Miocene shallow plutons and coeval volcanic rocks [*Swanson et al.*, 1989; *Evarts and Swanson*, 1994]. This belt contains, but is far wider than, the zone of large Quaternary stratovolcanoes that dominate the present landscape. Volcanism in the Cascade arc is considered discontinuous in space and time [*Guffanti and Weaver*, 1988] and is superposed on older Eocene rocks of the much broader and more diffuse Challis magmatic field [*Vance*, 1982]. The study area is also underlain by the southern Washington Cascades conductor (SWCC; Figure 1). Defined by magnetotelluric profiles, this high-conductivity anomaly dips eastward and is inferred to be an accreted marine sedimentary sequence, containing hypersaline fluids, within the compressed forearc basin [*Stanley et al.*, 1987].

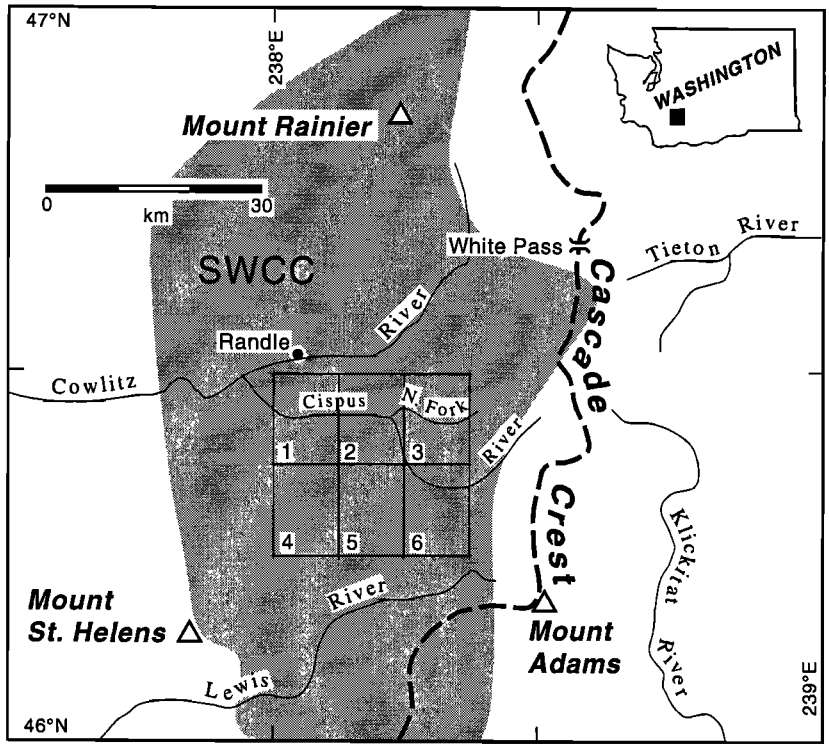


Figure 1. Map of southern Washington (see inset map) including the sampling area for this study (six 7.5' quadrangles) on the western flank of the Cascade arc between the Cowlitz and Lewis Rivers. The three late Pleistocene and Holocene composite volcanoes in southern Washington (Mount Rainier, Mount St. Helens, and Mount Adams) are also shown. Shaded region denotes part of the area underlain by the southern Washington Cascades conductor (SWCC).

Rocks of the Cascade arc have been divided by earlier workers into two major stratigraphic units: the Oligocene Ohanapecosh Formation which is overlain, apparently unconformably, by the uppermost Oligocene and lower Miocene Fifes Peak Formation [*Fiske et al.*, 1963; *Vance et al.*, 1987]. The Ohanapecosh Formation includes thick sequences of well-bedded volcaniclastic rocks having intermediate compositions and andesitic lava flows with minor silicic lava flows and ash flow tuffs. The Fifes Peak Formation consists mostly of andesitic to basaltic lava flows, pyroclastic rocks and lahars, and a few thick silicic ash flow tuffs. These deposits, defined near Mount Rainier National Park, cannot be readily traced through the area of this study, and so we do not apply these terms to rocks in our study area. Volcanic rocks in the 18 to 5 Ma range are sparsely represented in southern Washington, although scattered intrusions from this time interval are present (mainly the intrusive suite of Kidd Creek but perhaps others as well) [*Swanson et al.*, 1989].

Paleomagnetic data for Oligocene and Miocene rocks in western Washington and Oregon indicate that they have been rotated clockwise relative to the continental interior [*Magill et al.*, 1982; *Wells and Coe*, 1985; *Gromme et al.*, 1986]. The amount of rotation tends to increase with increasing age and also tends to increase toward the western plate margin [*Wells and Heller*, 1988]; Eocene oceanic basalt which forms the accreted basement of the Coast Ranges in Oregon and Washington is rotated as much as 80° clockwise [*Wells et al.*, 1985]. The youngest rotated rocks in the region are middle Miocene flood basalt of the Columbia River Basalt Group, which are rotated clockwise 16° to 30° relative to the same flows on the Columbia Plateau [*Wells et al.*, 1989]. The westward increase in observed rotations is consistent with distributed dextral shear between the North American and western oceanic plates during the late Cenozoic, although Basin and Range extension also played an important role [*Magill et al.*, 1982]. In their analysis of tectonic rotations in the Pacific Northwest, *Wells and Heller* [1988] concluded that dextral shear and extension of the Basin and Range province were subequal contributors to the total observed rotation.

3. Methods

3.1. Paleomagnetism

Paleomagnetic analyses of samples of the Kidd Creek dikes and sills were made to determine their characteristic directions of remanent magnetization. A common direction between the dikes and sills would support the hypothesis that these rocks are comagmatic. In addition, sites were selected (Figure 2) so that dikes and sills cutting well bedded host rocks with a range of attitudes would be sampled (Table 1). A fold test then could be used to determine whether the paleomagnetic directions for the dikes and sills are better grouped before or after application of structural corrections. A significantly better grouping of the paleomagnetic directions before unfolding would indicate that the intrusive suite of Kidd Creek postdates folding of the host rocks.

Oriented samples of the intrusive suite of Kidd Creek were collected at 12 sites in sills and 11 sites in dikes, using both magnetic and solar compasses. Areas of extensive alteration were avoided, and samples from only the freshest possible exposures were collected. Structural attitudes for host rocks are based on flow contacts or interbedded sediments where available. The remanent magnetization of individual specimens, cut

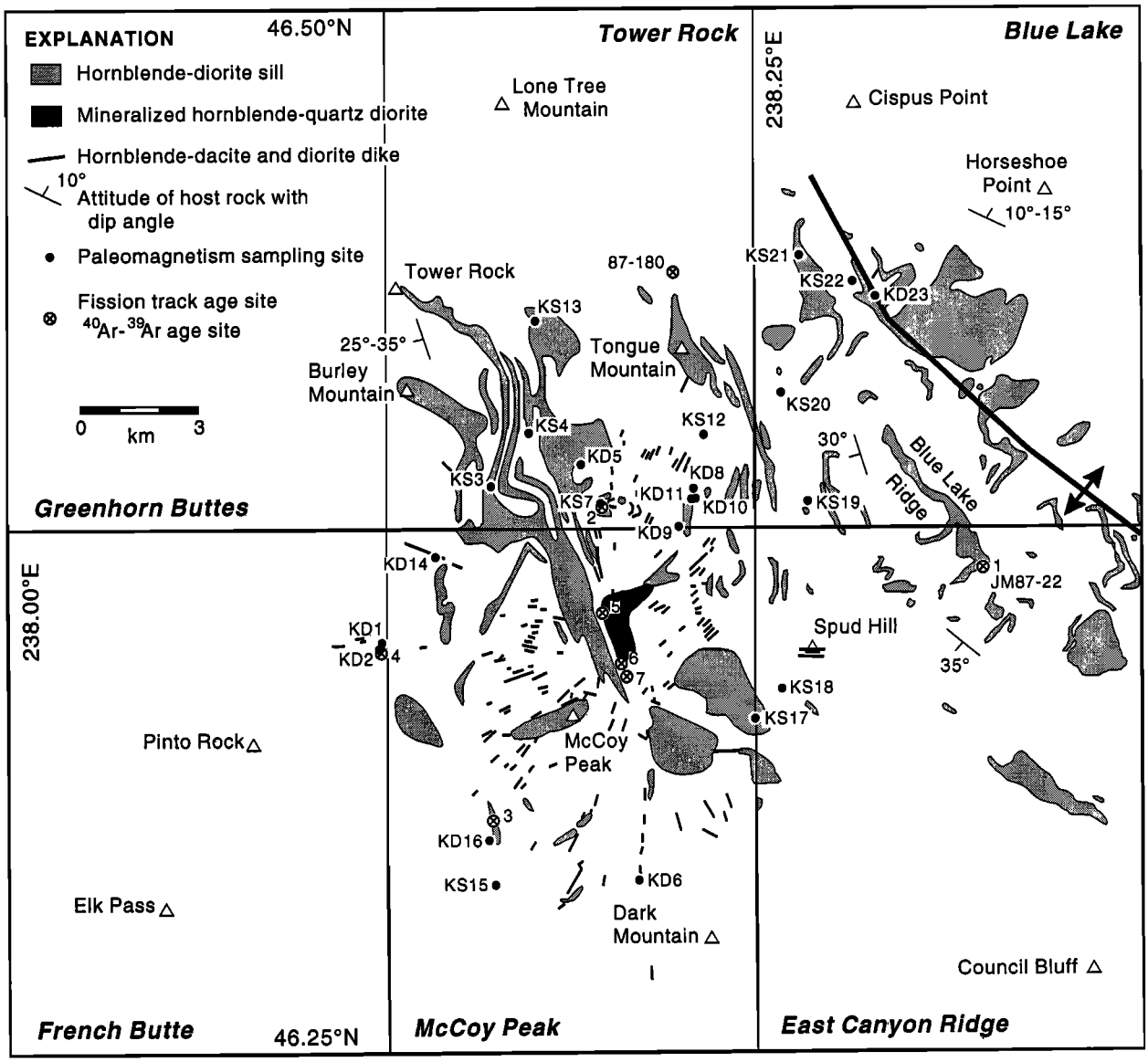


Figure 2. Generalized distribution map of the intrusive suite of Kidd Creek exposed within the six 7.5' quadrangles shown in Figure 1. Paleomagnetic sampling sites for this study are indicated by small dots with their corresponding site labels (Table 3). Sites of samples dated by the zircon fission track and ^{40}Ar-^{39}Ar methods are also shown by a circled cross (Tables 2 and 3). The axis of the Bishop Ridge anticline is shown by the thick NW-SE trending line in the Blue Lake quadrangle.

from the core samples, was measured with a cryogenic magnetometer. Specimens from all samples were progressively demagnetized with either standard alternating field (af) or thermal procedures. Lines representing the characteristic magnetization directions were fitted to the demagnetization data using a least squares method based on principal component analysis [*Kirschvink*, 1980], and the statistics of *Fisher* [1953] were used in analyzing the site and overall mean directions (Table 1).

3.2. Geochronology

Fission track dating of zircons from the intrusive suite of Kidd Creek and a wall rock sample was done by the external detector method as described by *Naeser* [1976]. The standard error in the fission track ages is a function of the number of tracks counted, and is estimated to be about 10% of the age (Table 2).

Hornblende phenocrysts for ^{40}Ar-^{39}Ar dating were irradiated in the TRIGA reactor at the U.S. Geological Survey in Denver, Colorado, for 30 hours at 1 MW, and variations in neutron flux were monitored with the MMhb-1 (K-Ar age of 520.4 ± 1.7 Ma) standard [*Samson and Alexander*, 1987]. Corrections for interfering isotopes were made using production ratios derived from potassium and calcium salts irradiated simultaneously with the samples [*Roddick*, 1983]. The samples were heated in a double-vacuum, low-volume resistance furnace in a series of 6 to 10 steps between 550° and 1375°C (Figure 3 and Table 3). Apparent ages for each temperature step were calculated using the decay constants recommended by *Steiger and Jäger* [1977]. Details of the analytical method similar to that used in this study are described by *Tysdal et al.* [1990].

4. Discussion of Results

Representative orthogonal plots of demagnetization data for the intrusive suite of Kidd Creek (Figure 4) typically show well-behaved single component magnetizations of normal po-

Table 1. Paleomagnetic Data for the Intrusive Suite of Kidd Creek, Southern Washington

Site	I_t	D_t	Strike/ Dip	I_c	D_c	N	R	k	α_{95}	λ_{VGP}	ϕ_{VGP}
KD1	68.0	26.6	140/6	73.3	18.7	7	6.9567	139	5.2	72.0	303.6
KD1	-68.6	203.7	140/6	-73.7	194.7	6	5.8932	47	9.9	73.6	299.9
KD2	68.5	12.4	140/6	72.8	0.7	8	7.9199	87	6.0	80.3	290.4
KS3	61.3	9.1	160/25	63.3	319.2	9	8.8712	62	6.6	82.4	356.7
KS4	44.3	28.1	160/32	61.0	349.7	9	8.9572	187	3.8	59.8	0.9
KD5	64.0	12.6	155/35	62.8	294.5	9	8.9540	174	3.9	81.2	328.0
KD6	56.8	9.5	160/25	60.8	327.1	7	6.9378	96	6.2	78.6	16.3
KS7	70.2	11.0	165/27	64.9	301.0	9	8.8521	54	7.1	79.5	276.0
KD8	66.6	354.1	150/30	61.2	288.7	8	7.9006	70	6.7	85.2	185.1
KD8	-64.8	206.9	150/30	-74.2	118.4	8	7.8666	52	7.7	71.6	317.4
KD9	66.6	15.3	160/35	62.1	293.7	8	7.8533	48	8.1	79.4	307.4
KD10	43.0	21.3	150/30	61.2	348.6	10	9.9165	108	4.7	62.7	12.4
KD11	16.6	92.7	150/30	40.6	103.0	10	2.4932	1	-.-	4.3	320.4
KS12	65.5	22.7	150/33	70.5	289.0	8	7.7566	29	10.5	74.6	315.3
KS13	65.1	356.3	165/40	48.4	293.5	10	9.7164	32	8.7	87.4	165.2
KD14	65.6	12.9	140/15	74.8	338.3	9	8.9249	107	5.0	81.1	314.3
KS15	67.6	51.7	50/26	55.7	97.4	11	10.9336	151	3.7	56.1	301.9
KD16	67.2	27.5	160/15	74.7	348.1	8	7.9222	90	5.9	71.4	306.8
KS17	-49.5	243.3	130/35	-74.8	298.4	9	8.9175	97	5.3	39.3	323.2
KS18	70.5	14.7	150/85	18.5	254.5	10	9.8950	86	5.3	77.5	281.1
KS19	65.3	2.6	170/33	54.3	304.2	9	8.9412	136	4.4	88.0	296.2
KS20	67.1	7.7	110/40	71.8	215.4	9	8.9471	151	4.2	83.8	291.5
KS21	68.8	7.6	330/23	50.6	33.2	10	9.9457	166	3.8	82.4	276.0
KS22	-55.2	233.3	280/32	-28.5	216.4	10	9.9653	260	3.0	49.3	323.8
KD23	81.9	8.6	285/33	49.0	13.6	7	6.7982	30	11.3	62.0	243.3
$Mean_t$	66.6	18.1				23	22.6334	60	3.9	77.6	309.2
$Mean_c$				72.1	333.1	23	20.5243	9	10.8	70.4	191.1

Site label, D indicates dike and S indicates sill; I_t and D_t are in situ inclination and declination (degrees) of paleomagnetic directions (I_c, D_c, corrected for structural tilt: Strike/Dip, strike and dip of host rock, strike 90° counterclockwise of dip direction); N is number of samples (sites) analyzed; R is vector sum of N unit vectors; k is the concentration parameter [*Fisher*, 1953]; α_{95} is the radius of 95% confidence (degrees); λ_{VGP} and ϕ_{VGP} are latitude and east longitude of virtual geomagnetic pole (degrees) uncorrected for structural tilt. Sites KD8 (recent normal overprint) and KD11 (probably lightning struck) were not included in the overall means.

larity (e.g., Figures 4b and 4c and Table 1). In general, within-site directions of natural remanent magnetization (NRM) for samples measured before demagnetization are well grouped. For two sites (KS17 and KS22), however, the single component of magnetization observed is of reversed polarity (Figure 4e) indicating these sills were emplaced under different field conditions. Samples from two other sites (KD1 and KD8) have two components of magnetization, one normal and one reversed.

The characteristic reversed-polarity direction of site KD8 is overprinted by a normal-polarity component that is similar to the present-day direction of the geomagnetic field (Figure 4d). Recent overprint components are present in samples from other sites as well, although they are not as well defined. The antipodal normal- and reversed-polarity directions observed in individual samples from site KD1 (Figure 4a) indicate that this dike (8 m thick) was initially emplaced in a normal-polarity field. The reversed-polarity overprint probably results from

Table 2. Zircon Fission Track Ages for the Intrusive Suite of Kidd Creek and Its Wall Rock, Southern Washington

Map	Sample	Unit	Number of fossil tracks	Number of induced tracks	Fluence	Age
1	JM87-22	Kidd Creek	635	1561	0.94×10^{15}	11.4±1.1
2	DS85-34	Kidd Creek	328	900	0.95×10^{15}	10.4±1.0
3	DS85-51	Kidd Creek	238	506	0.95×10^{15}	13.4±1.3
4	DS86-138	Kidd Creek	315	720	0.97×10^{15}	12.7±1.3
5	DS90-126	Kidd Creek	483	1412	0.97×10^{15}	10.0±1.0
6	DS90-127	Kidd Creek	394	998	0.97×10^{15}	11.5±1.2
7	DS91-95	wall rock	1498	2406	1.00×10^{15}	18.7±1.9
						11.6±1.3*

Age reported with estimated 10% error; 1 to 3 counted by J. A. Vance; 4 to 7 counted by D. A. Swanson; 2 is from paleomagnetic site KS7; and 4 is from paleomagnetic site KD2.

*Mean fission track age for intrusive suite of Kidd Creek.

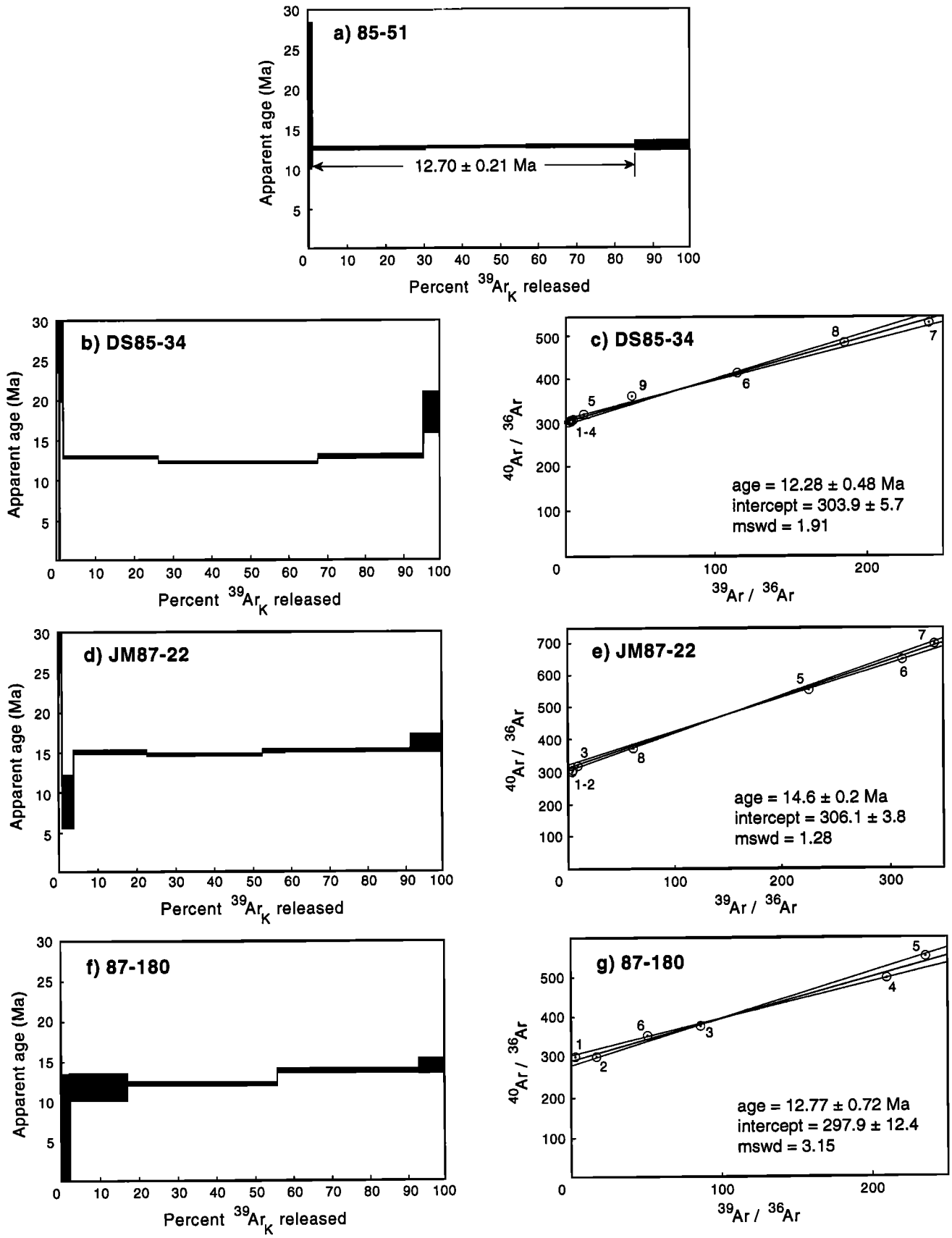


Figure 3. The ^{40}Ar-^{39}Ar hornblende age spectra and isochrons for samples from the intrusive suite of Kidd Creek (Table 3). For the age spectra (Figures 3a, 3b, 3d, and 3f), the vertical width of each bar represents the analytical error in the age of the individual heating step at $\pm 1\sigma$. For the isochons (Figures 3c, 3e, and 3g), the numbered points indicate the heating steps, and the fine lines the 2σ error in the isochon's slope.

Table 3. The ^{40}Ar-^{39}Ar Data for Mineral Separates From Dikes and Sills of the Intrusive Suite of Kidd Creek, Southern Washington

Temperature °C	Radiogenic ^{40}Ar	K-Derived ^{39}Ar	$^{40}Ar_R/^{39}Ar_K$	$^{39}Ar/^{37}Ar$	Radiogenic Yield, %	Percent ^{39}Ar	Apparent Age and Error, Ma
			*85-51; Hornblende**				
550	0.0037	0.0004	10.01	0.59	5.0	0.0	121±17
650	0.0003	0.0004	0.72	0.08	0.5	0.1	9±14
750	0.0068	0.0006	10.62	0.09	1.5	0.1	129±32
900	0.0019	0.0013	1.50	0.09	1.7	0.2	18.8±1.9
1050	0.0072	0.0046	1.56	0.06	5.3	0.6	19±9
1125	0.0850	0.0842	1.01	0.00	25.3	11.3	12.61±0.13
1150	0.1370	0.1354	1.01	0.05	32.3	18.2	12.65±0.12
1200	0.2033	0.1990	1.02	0.04	33.0	26.7	12.78±0.04
1250	0.2143	0.2113	1.04	0.05	28.7	28.3	12.69±0.18
1350	0.1107	0.1086	1.02	0.05	19.0	14.6	12.75±0.60
			DS85-34; Hornblende†				
700	0.0052	0.001	3.9	0.18	2.2	0.3	48±16
850	0.0076	0.0016	4.85	0.21	1.0	0.4	59±5
950	0.0019	0.001	1.8	0.06	1.5	0.2	23±34
1025	0.0056	0.001	4.42	0.03	5.2	0.3	57±33
1075	0.0086	0.0039	4.46	0.04	7.5	0.9	27±7
1125	0.10807	0.10332	4.46	0.04	28.7	24.5	12.91±0.19
1175	0.17270	0.17371	4.45	0.04	44.7	41.2	12.3±0.12
1250	0.12277	0.11632	4.46	0.04	39.7	27.6	59.75±0.29
1350	0.0284	0.0188	4.47	0.05	18.4	4.5	59.83±2.6
			JM87-22; Hornblende‡				
700	0.0072	0.0006	11.4	0.22	2.5	0.1	140±42
950	0.0198	0.003	7.6	0.14	4.6	0.5	49.4±0.3
1050	0.0089	0.003	3.5	0.07	7.0	0.5	52.24±0.10
1100	0.0119	0.017	0.70	0.06	10.0	3.1	52.87±0
1150	0.12479	0.10524	1.186	0.06	47.4	19.0	52.23±0.26
1200	0.19231	0.16533	1.163	0.06	54.9	29.9	52.5±0.5
1250	0.25847	0.21425	1.206	0.06	58.1	38.7	52.63±0.16
1350	0.0586	0.04567	1.28	0.06	20.7	8.3	52.7±0.4
			87-180; Hornblende§				
1000	0.0022	0.0008	2.99	0.07	1.9	0.2	37±19
1100	0.0013	0.0068	0.20	0.05	1.1	2.2	2.5±11
1150	0.0445	0.0474	0.94	0.04	21.5	15.0	11.9±1.7
1200	0.11877	0.12276	0.968	0.04	40.7	38.9	12.21±0.24
1250	0.12740	0.11651	1.093	0.04	46.5	36.9	13.79±0.29
1375	0.0246	0.0216	1.139	0.05	16.5	6.8	14.4±0.9

Temperature is of the heating step. Radiogenic ^{40}Ar and K-derived ^{39}Ar abundances reported in volts, calculated to five decimal places and rounded to significant figures using analytical precisions; conversion to moles can be made using 9.74×10^{-13} mol argon per volt of signal. Detection limit at the time of this experiment was 2×10^{-17} mol argon. The $^{40}Ar_R/^{39}Ar_K$ ratio is calculated to three decimal places and rounded to significant figures using analytical precisions and has been corrected for all interfering isotopes including atmospheric argon. Mass discrimination was determined by measuring the $^{40}Ar/^{36}Ar$ ratio of atmospheric argon; our measured value is 298.9 during the period of this experiment, and the accepted ratio is 295.5. Abundances of interfering isotopes of argon from K and Ca were calculated from reactor production ratios determined by irradiating and analyzing pure CaF_2 and K_2SO_4 simultaneously with these samples. The measured production rations for these samples are $(^{40}Ar/^{39}Ar)_K = 6.20 \times 10^{-3}$, $(^{38}Ar/^{39}Ar)_K = 1.308 \times 10^{-2}$, $(^{37}Ar/^{39}Ar)_K = 1.27 \times 10^{-4}$, $(^{36}Ar/^{37}Ar)_{Ca} = 2.71 \times 10^{-4}$, $(^{39}Ar/^{37}Ar)_{Ca} = 5.05 \times 10^{-4}$, and $(^{38}Ar/^{37}Ar)_{Ca} = 1.80 \times 10^{-5}$. Corrections were also made for additional interfering isotopes of argon produced from irradiation of chlorine using the method of *Roddick* [1983]. The $^{39}Ar/^{37}Ar$ ratio can be divided by 2 to calculate approximate K/Ca ratios. Apparent ages and associated errors were calculated from unrounded analytical data and then rounded using associated errors; 1σ error reported for raw data; error for plateau and isochron dates is 2σ. Decay constants are $\lambda_e = 0.581 \times 10^{-10} yr^{-1}$ and $\lambda_\beta = 4.962 \times 10^{-10} yr^{-1}$ [*Steiger and Jäger*, 1977]. Uncertainties for the date of individual steps in a spectrum were calculated using modified equations of *Dalrymple et al.* [1981]. J value, neutron flux parameter calculated from MMhb-1 irradiation monitor with percent K = 1.555, $^{40}Ar_K = 1.624 \times 10^{-9}$ mol/g, and K-Ar age = 520.4 Ma [*Samson and Alexander*, 1987]. JM87-22 is from the same sample as in Table 2; DS85-34 is from paleomagnetic site KS7; and 85-51 is from paleomagnetic site KS15.

*Total gas date, 12.9 ± 0.2 Ma; plateau date, 12.70 ± 0.21 Ma (1125°-1350°C); J = 0.006956 ± 0.1%; weight = 219.2 mg.

†Total gas date, 13.49 ± 0.39 Ma; no plateau; isochron date, 12.28 ± 0.48 Ma (all); $(^{40}Ar/^{39}Ar)_i = 304 \pm 3$; J = 0.006864 ± 0.1%; weight = 150.2 mg.

‡Total gas date, 15.66 ± 0.47 Ma; no plateau; isochron date, 14.6 ± 0.2 Ma (all excluding 1100° step); $(^{40}Ar/^{39}Ar)_i = 306 \pm 2$; J = 0.007076 ± 0.1%; weight = 150.2 mg.

§Total gas date, 12.74 ± 1.7 Ma; no plateau; isochron date, 12.77 ± 0.72 Ma (all); $(^{40}Ar/^{39}Ar)_i = 292 \pm 6$; J = .007019 ± 0.1%; weight = 100.0 mg.

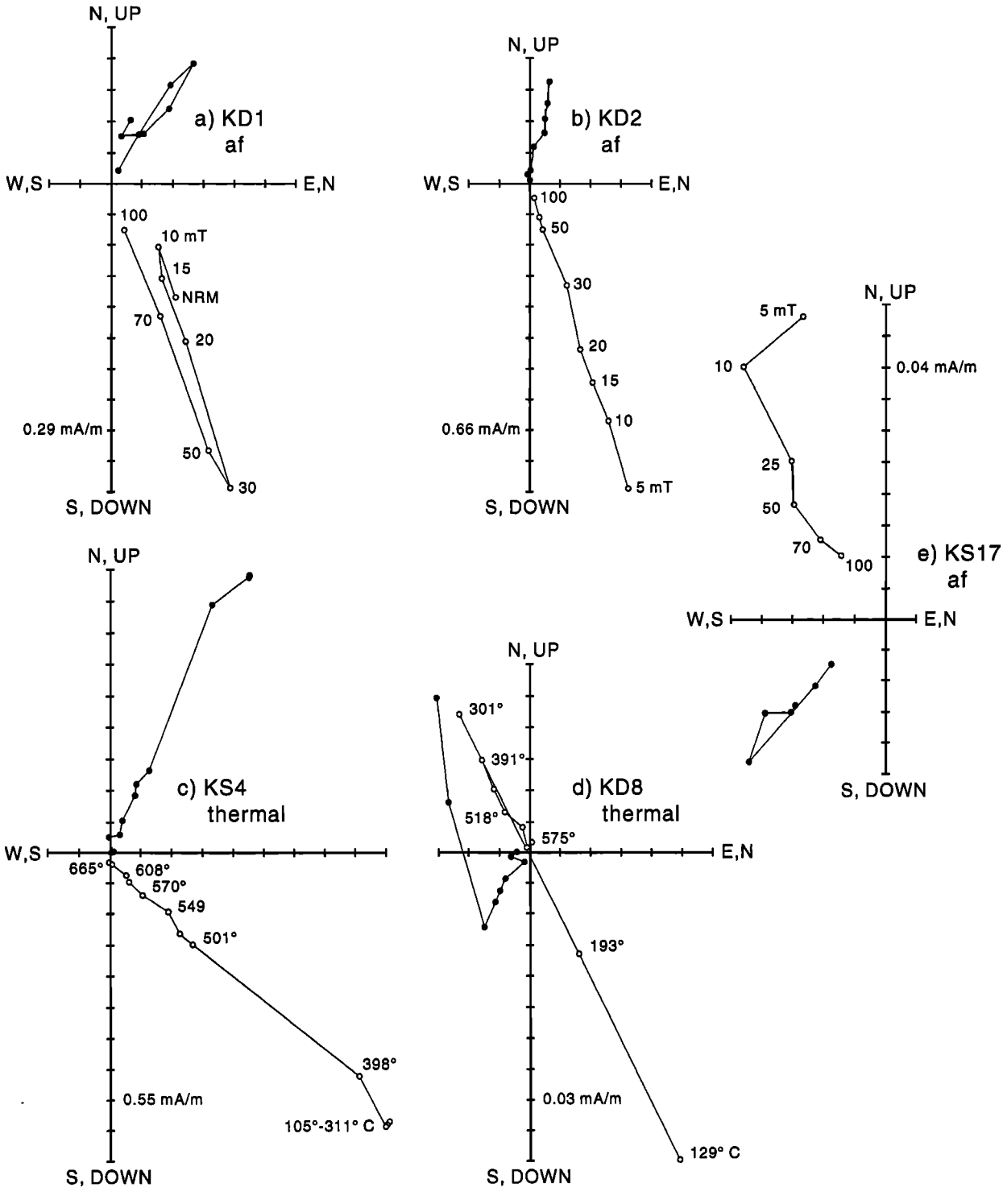


Figure 4. Orthogonal projections (uncorrected for tilt) of alternating field (af) and thermal demagnetization vector endpoints for the Kidd Creek samples. Solid circles indicate projections onto the horizontal plane, and open circles indicate projections onto the vertical plane.

reheating associated with the nearby emplacement of a dike or sill after a polarity reversal. Randomly dispersed NRM and characteristic directions [*Watson*, 1956] for site KD11, from a ridge crest exposure, indicate that these samples most likely have been affected by lightning-induced magnetizations.

Thermal and af demagnetization results are generally similar for samples from a given site and indicate that the remanent magnetization for most samples is carried by magnetite (e.g., Figure 4d). However, thermal demagnetizations to 670°C show that samples from at least four sites also contain hematite with the same remanence direction as in magnetite (e.g., Figure 4c). In some cases, the demagnetization data indicate the presence of a second high-temperature, high-coercivity, and reversed-polarity component of magnetization at the highest demag-

netization steps. These trajectories, however, are of poor quality and could not be fitted with either least squares lines or planes.

Site mean directions for most of the Kidd Creek dikes and sills are of normal polarity and, uncorrected for structural tilt (Figure 5a), are statistically the same at the 95% confidence level [*McFadden and Jones*, 1981]. Although statistically different from the overall mean, site means KS15 and KD23 are considered part of this group; the dike at site KD23 was the most hydrothermally altered of the rocks sampled. Reversed-polarity directions for sites KS17 and KS22 have means that are clearly distinct, although those for sites KD1 and KD8 are not, and their antipodes have been plotted for comparison with the other normal-polarity directions. Site means KS4 and KD10 also indicate that these rocks were emplaced at some other time. The differences in direction between these groups of site means (Figure 5a) are most likely due to paleosecular variation of the geomagnetic field. The angular standard deviation value (S) for the Kidd Creek VGPs (Table 1) is 15.6°. This value is identical to the lower 95% confidence limit on a model value of S_f = 16.7° (S_L= 15.6°, S_u= 18.0°) for the 40° to 50° latitude range (λ_m= 45.1°) based on paleomagnetic data for lava flows over the last 5 m.y. [*McFadden et al.*, 1988]. This value might be low because the individual dikes and sills cooled over a longer period of time and likely averaged some PSV compared to the lava flows.

Although the fission track and ^{40}Ar-^{39}Ar results range between 10 and 15 Ma (Tables 2 and 3), there is no demonstrable geologic significance to these age differences. We place the highest confidence in the ^{40}Ar-^{39}Ar plateau age from sample 85-51 of 12.70 ± 0.21 Ma, which is a minimum age of emplacement. The other three samples show excess argon as evidenced by the age spectra and isochrons. The result for sample 87-180 corroborates the 12.70 Ma plateau age, and the result for DS85-34 is the same within analytical error. The oldest ^{40}Ar-^{39}Ar age of 14.6 ± 0.2 Ma from sample JM87-22 suggests an older (maximum) age of emplacement but also has a fission track age of 11.4 ± 1.1 Ma. Part of this difference could be that the fission tracks formed at a temperature of ~200°C, whereas the ^{40}Ar-^{39}Ar age records an earlier stage in the emplacement; perhaps the fission track age is less accurate. The mean fission track age of all samples is 11.6 ± 1.3 Ma. Clearly, the normal- and reversed-polarity paleomagnetic data indicate that the Kidd Creek rocks span at least one polarity reversal, and the simplest interpretation is that the normal-polarity rocks were emplaced within a single normal-polarity chron during a relatively brief interval of time.

5. Conclusions

The overall similarity of the in situ site-mean directions for the intrusive suite of Kidd Creek (Figure 5; Table 1) indicates that these rocks are most likely comagmatic, which corroborates the geochemical data of *Marso and Swanson* [1992]. The site-mean directions are plotted after correction for structural tilt of the host rocks in Figure 5a. Compared to in situ data (Figure 5b) the overall dispersion has increased, and the concentration parameter k [*Fisher*, 1953] has decreased from 60 to 9. Thus the paleomagnetic directions for the Kidd Creek rocks fail a fold test at the 99% confidence level [*McElhinny*, 1964],

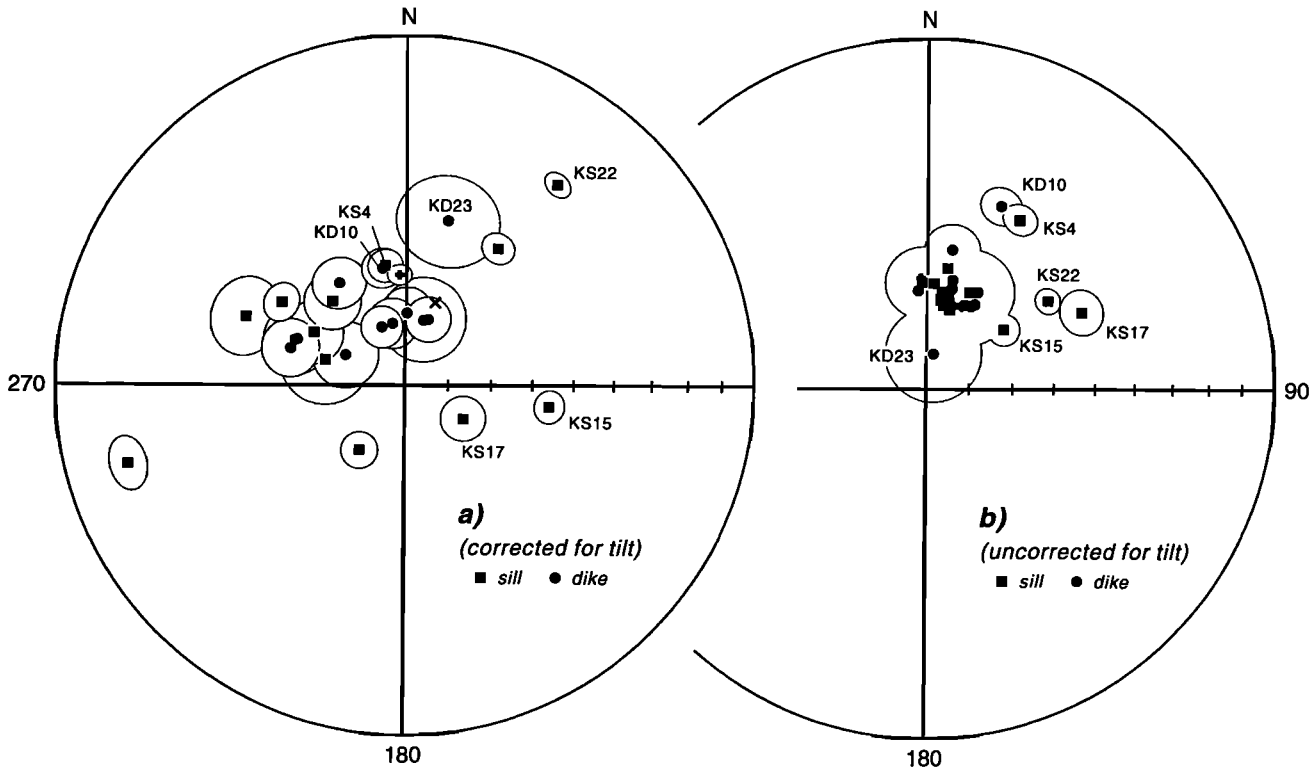


Figure 5. Equal-area plots of site-mean directions with their 95% confidence limits for the Kidd Creek samples. Solid symbols indicate lower hemisphere points. Directions in Figure 5a are corrected for structural tilt, those in Figure 5b are in situ. Normal-polarity antipodes are plotted for reversed-polarity site means (Table 1). Pluses and crosses indicate expected mid-Miocene and present-day field directions, respectively, for southern Washington. The site-mean directions fail a fold test at the 99% confidence level indicating that the intrusive suite of Kidd Creek postdates regional folding (see text).

indicating that they were emplaced after folding had occurred. The site-mean directions KS17 and KS22 diverge after structural correction indicating that these sills also are postfolding. The mean in situ direction also indicates that the Kidd Creek rocks are rotated 22° ± 6° clockwise about a vertical or near-vertical axis relative to the expected Miocene direction for the region [*Hagstrum et al.*, 1987]. Our result is consistent with rotations observed in Miocene flows of the Columbia River Basalt Group from the Coast Ranges of Washington and Oregon [*Wells et al.*, 1989] and indicates that younger rocks of the Cascade arc farther east were also affected. Oligocene and Eocene rocks of the western Cascades in Washington and Oregon had previously been shown to be rotated clockwise between ~20° and ~35° [*Bates et al.*, 1981; *Magill and Cox*, 1980; *Beck and Burr*, 1979].

The paleomagnetic data for the Kidd Creek dikes and sills indicate that these rocks were emplaced after termination of local compressional deformation over an interval in time that included at least one reversal of the geomagnetic field. We infer the general sequence of events to have occurred as follows: (1) cessation of compression, (2) emplacement of the bulk of Kidd Creek intrusions in a normal-polarity field, (3) emplacement of sills KS17 and KS22 before, in between, or after emplacement of the other Kidd Creek rocks in a reversed-polarity field, (4) overprinting of dike KD1 in magnetite and other rocks in hematite in this or a different reversed-polarity field, (5) clockwise vertical-axis rotation in the region of ~22°, and (6) recent acquisition (normal-polarity field) of thermoviscous components in dike KD8 and other Kidd Creek rocks. Assuming the intrusive suite of Kidd Creek was emplaced at ~12.7 Ma, our data indicate that compression and regional folding in southern Washington had ended by this time prior to the onset of rotational deformation. An additional zircon fission track age of about 19 Ma (Table 2, map 7) for a quartz-phyric ash flow tuff in tilted wall rock of the McCoy Creek intrusion indicates that folding occurred between ~19 and 12 Ma.

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